

PRANJAL KAR

AI/ML Engineer – Generative AI & AI Agents

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PROFESSIONAL SUMMARY

AI/ML Engineer with 2.5+ years of experience owning end-to-end development of production AI systems and LLM-powered agents. Proven expertise in building, deploying, and optimizing AI agents using LangChain, LlamaIndex, and fine-tuned transformer models. Strong product mindset with hands-on experience integrating AI into user-facing applications, improving data pipelines, and collaborating cross-functionally in fast-paced startup environments.

CORE SKILLS

- **Generative AI:** LLM Agents, LangChain, LlamaIndex, RAG Systems, Prompt Engineering.
- **Machine Learning:** Fine-tuning Transformers (BERT, GPT), PyTorch, TensorFlow, Hugging Face.
- **Data Engineering:** Vector Databases, Elasticsearch, Embeddings, Kafka, PySpark.
- **MLOps & Infrastructure:** Docker, Kubernetes, MLflow, Redis, Celery.
- **Backend & Cloud:** Python, FastAPI, Flask, GCP (Cloud Run), AWS.

PROFESSIONAL EXPERIENCE

Project Engineer – Machine Learning Jan 2024 – Present
Centre for Development of Advanced Computing (CDAC) Bangalore, India

- Owned end-to-end development of AI-driven security systems from model design to production deployment.
- Architected and deployed deep learning pipelines processing 1M+ events per hour with 95% accuracy.
- Improved anomaly detection performance by 30% through model experimentation and architectural optimization.
- Enhanced data pipelines using Kafka, PySpark, and Redis to improve data quality and availability.
- Integrated AI models into backend systems using FastAPI and Flask, ensuring seamless interaction with core platforms.
- Monitored system performance, debugged failures, and continuously optimized models and pipelines.

Junior MLOps / AI Engineer (Freelance) Nov 2023 – Jan 2024
Ovvy Remote (USA)

- Built and maintained AI systems for deploying custom multimodal models as APIs.
- Designed **agent-like AI workflows** for task-specific inference and decision logic.
- Implemented prompt optimization and inference tuning, improving response latency by 40%.
- Deployed containerized AI services using Docker with scalable backend architectures.

Machine Learning Engineer – Generative AI Jan 2023 – Dec 2023
Compfox (Legal Tech Startup) Remote (USA)

- Built **LLM-powered agentic systems** for legal document analysis and drafting using **LangChain**, **LlamaIndex**, and structured tool-calling.
- Designed and implemented **RAG pipelines** over 20K+ legal documents using **embeddings**, **semantic chunking**, and **vector retrieval** (Elasticsearch/Pinecone).
- Fine-tuned **transformer NLP and Llama models** for classification, clause similarity, and recommendation tasks on private medical docs.
- Deployed **containerized GenAI services** on **GCP Cloud Run** with **FastAPI**, **vector DBs**, and orchestration layers.
- Engineered **multi-step agent workflows** (*retriever → reasoner → verifier → drafting*) with HITL feedback and deterministic guardrails.

EDUCATION

Indian Institute of Technology, Madras 2021 – 2024
Diploma in Data Applications (Data Science)

Tezpur University 2020 – 2024
Bachelor of Technology – Electronics & Communication Engineering CGPA: 8.4 / 10.0

KEY PROJECTS

- **Shruti-Drishti – Multimodal AI Agent for Sign Language:** Built a real-time ISL detection system (CNN+LSTM) with LLM-based English ↔ ISL translation via LangChain. Winner of Hasgeek Hackathon.
- **Network Intrusion Detection System:** Designed AI system processing 1M+ packets/hour using ensemble models and distributed architecture (Kafka, PySpark, Flask).

CERTIFICATIONS & AWARDS

- **Winner** – Open Source AI Hackathon, Fifth Elephant 2024
- **Deep Learning Specialization** – Andrew Ng | **Machine Learning Specialization** – Stanford
- **Google Cloud Professional ML Engineer** – In Progress